

Jay Patel

Toronto, Ontario | 2594jaypatel@gmail.com | +1 437-662-7814 | [GitHub](#) | [LinkedIn](#)

EDUCATION

York University

BSc (Hons.) Computer Science

Toronto

Expected: May 2028

- Relevant Coursework:** Advanced Math(Calculus, 3D Geometry, Probability, Statistics), Programming (Python, SQL, Java, C), Data Structures and Algorithms, Artificial Intelligence, Relational databases, Risk Management

EXPERIENCE

Royal Bank of Canada (RBC)

Data Developer Co-op

Toronto

May '26-Present

- Developed a FinBERT NLP model to map unstructured fraud descriptions to standardized risk levels, outperforming a XGBoost baseline by 17%.
- Automated risk hierarchy classification across 10k+ transit with Pandas pipeline and reduced manual time by 90%.
- Designed SQL ETL queries to resolve data anomalies and map historical risk logs into a unified schema.
- Performed data transformation using Azure Data Factory and Databricks on large data sets.
- Leveraged Spark, Hadoop and Airflow to integrate data from various sources.
- Created dashboards in Power BI and Dash to visualize hierarchical and risk data for cross-departmental teams.

York University

Research (ACM FSE Conference 2026)

Toronto

Sep. '25-Dec. '25

- Designed a Reinforcement Learning system using PyTorch, Sentence Transformers, and FastAPI.
- Applied NLP and transformer based reasoning to guide RL rewards, improving code generation accuracy by 15% on 400+ competitive programming problems.
- Analyzed 300+ GitHub logs in Microsoft Excel for faculty research, identifying software bugs.

PROJECTS

Open-Source Contributor - Pandas (10 Million+ global users) | [GitHub](#)

- Bug fix and added test cases for ExtensionArray operations; 15+ commits with multiple code reviews.
- Improved CSV parsing with on_bad_lines; contributed to iterative development, unit testing and debugging.

Demand Forecasting and Anomaly Detection Dashboard | [GitHub](#)

May 2025

- Processed 40M+ NYC taxi trips, cleaned 1.2M+ outliers, and added new features using PySpark, Dask, and Pandas.
- Trained predictive models - XGBoost (13% MAPE) for demand and LSTM (0.055 MSLE) for detecting anomalies.
- Built a Power BI dashboard from model outputs stored in MySQL for easy visualization and analysis.

Skills: Data Engineering, Power BI, SQL, Python, XGBoost, LSTM, PySpark, Dask, Pandas, Data Mining, Tensorflow

Quantitative Trading Algorithm with LSTM | [GitHub](#)

July 2025

- Built an ensemble trading model with LSTM, Bollinger Bands, and RSI, executing trades when 2+ indicators aligned.
- Achieved +54.6% cumulative return and 6.5% CAGR over 5 years across 50+ stocks.
- Outperformed the S&P 500 by 17%, achieved a 57% return in the best year.

Skills: Time Series Forecasting, Feature Engineering, Statistical Analysis, LSTM, Tensorflow, NumPy, scikit-learn, Data Modeling

AI RAG Assistant Using LangChain + LLMs | [GitHub](#)

September 2025

- Built a context-aware chatbot using LangChain and Watsonx LLM to provide answers from documents.
- Created a document data pipeline with RAG that splits PDFs into 100-token chunks.

Skills: IBM Watsonx LLM, LangChain, Retrieval-Augmented Generation (RAG), Semantic Search, Prompt Engineering

TECHNICAL SKILLS

Machine Learning Algorithms: Regression, Classification, Clustering, Transformers, LSTM, Reinforcement Learning

LLM & NLP Frameworks: IBM Watsonx, LangChain, LangGraph, Hugging Face, RAG, Open AI API

Libraries: scikit-learn, TensorFlow, Keras, PyTorch, NumPy, Pandas, PySpark, XGBoost, JAX, Matplotlib, Plotly

Languages: Python, SQL, R, Scala, Java

Cloud & DevOps: Docker, Kubernetes, Azure, GCP, CI/CD, FastAPI, Linux, Unix, MLflow, Git, Github

Other tools: Databricks, Datameer, Teradata, Tableau, Excel, PowerBI

CERTIFICATIONS

IBM AI Engineering Professional Certificate

- Built and deployed Generative AI and Deep Learning models with end-to-end ML workflows, including data analytics, preprocessing, feature engineering, model training, evaluation and cloud deployment workflows.